



COMSATS University Islamabad, Abbottabad Campus

Department of Computer Science

Lab Assignment 1 (Marks: 10)

Data Structures

Class: BCS- 3 A & B

- Q1. Implement a stack using array Data Structure that may supports two basic stack operations like push() and pop(). Also, demonstrate **stack overflow** and **underflow** conditions. Implement a **menu-driven program** for stack operations.
- Q2. Write a program that reads in a sequence of characters (String) and prints them in reverse order. Additionally write a function to **count total elements in the stack**.
- Q3. Write a program that reads in a positive integer and prints the binary representation of that integer by using stack.
- Q4. Write a program that checks whether an input string is a palindrome or not using a stack.
- Q5. Write a program to push five values into a stack. Pop values from the stack one by one, calculate the factorial of each value, and display the results.
- Q6. Write a program of stack to display the elements in FIFO manner.
- Q7. Implement a dynamic stack using a singly linked list. Write a program to push and pop elements dynamically (no size limit) and count number of nodes in the stack.
- Q8. Create a Doubly Linked List and do as directed

Sort the Doubly Linked List

- Sort elements using Bubble Sort or Insertion Sort.

Remove Duplicates

- Delete duplicate values from list.

Merge Two Doubly Linked Lists

- Combine two sorted lists into one.

Split the List

- Divide list into two halves.

Q9. Create a Circular Linked List

- Create nodes where last node points back to head.

Display the List

- Traverse from head and stop when it reaches head again.

Insert at Beginning

Add node before head and update last node link.

Insert at End

Insert node before head (last position).

Insert at Position

Insert at any given index.

Delete from Beginning

- Remove head and update last node.

Delete from End

- Remove last node and link second last to head.

Delete by Value

Remove node containing a specific value.

Count Nodes

- Count nodes using loop until head repeats.

Find Maximum & Minimum

- Traverse and compare values.

Check Circular Nature

- Verify if list is circular or not.

Convert Singly LL to Circular LL

- Modify last node pointer.